

check-workable-items.sh

Constitution rule: §11.4.93 (workable-items SQLite single-source-of-truth) + §11.4.95 (DB tracked-in-git, never gitignored). Gate ID: CM-WORKABLE-ITEMS-SYNC.

Type: Local CI gate (invoked by scripts/verify-all-constitution-rules.sh).

Last verified: 2026-08-26 (corpus-scope floor — both `_Summary` trackers are now regenerated from the DB and byte-compared; LVA vacuous-pass sweep F17)

What it does

Verifies Lava's workable-items tracker is internally consistent, using the **canonical** workable-items Go binary shipped in the constitution submodule (constitution/scripts/workable-items/), keyed LVA-N:

1. Builds the canonical binary if absent (from inside the module dir; `CGO_ENABLED=1` is required — cgo sqlite driver).
2. `workable-items validate --db docs/workable_items.db` — closed-set + §11.4.91 invariants, **filtered against the exemption ledger** at docs/superpowers/specs/2026-08-20-workable-items-evidence-exemptions.md (see "Exemption ledger" below). A violation whose (atm_id, history_id) is not listed in the ledger's machine-readable block still fails the gate.
3. `workable-items diff --db docs/workable_items.db --issues docs/Issues.md --fixed docs/Fixed.md` — exits 1 on DB↔Markdown divergence (CM-WORKABLE-ITEMS-MD-DB-IN-SYNC).
4. **Summary-tracker byte comparison (added 2026-08-26).**
Regenerates `Issues_Summary.md` + `Fixed_Summary.md` from the *same* DB into a temp dir and compares them byte-for-byte against the committed files. See "Corpus-scope floor" below.
5. `git ls-files --error-unmatch docs/workable_items.db` — asserts the DB is git-tracked (§11.4.95 CM-WORKABLE-ITEMS-DB-TRACKED).

Replaced the retired CM-LVA-TICKETS-SYNC gate (bespoke tools/lava-tickets/ system, migrated 2026-05-31 per docs/tickets/MIGRATION-TO-CANONICAL.md).

Exemption ledger — added 2026-08-20

67 pre-existing `validate` violations (dated 2026-06-09, predating this gate's strict wiring) have no honest fabrication-free fix: 49 point at evidence files bulk-deleted by commit 56f2417795c9 with the `workable-items` binary offering no "mark historically deleted, narrative stands" mechanism; 2 (same item, two history rows) name a malformed path that never existed under either its literal or `..-normalized` form; 16 have closure narratives naming a test class/module but no single unambiguous still-existing file (only bare module directories match — using one as "evidence" would be a guess). Two further violations (LVA-013 history ids 202/203) genuinely had two still-existing, unambiguous evidence files embedded in their own narrative and were `FIXED` via `correct-history-evidence`, not exempted.

Per the same §6.D Behavioral Coverage Contract precedent as docs/superpowers/specs/2026-04-30-sp3a-coverage-exemptions.md: every exemption is a named, dated, individually-investigated row — never a blanket waiver. `LAVA_WORKABLE_ITEMS_EXEMPTIONS` overrides the ledger path for hermetic testing. Full investigation + per-category reasoning: docs/superpowers/specs/2026-08-20-workable-items-evidence-exemptions.md.

Falsifiability rehearsal (this session): removed the LVA-009|16 line from the ledger → gate correctly `FAILED` with `CM-WORKABLE-ITEMS-SYNC`: `unexempted validate violation(s)`; restored the line → gate correctly `PASSED` again. docs/workable_items.db was independently confirmed unaffected by the rehearsal (`git diff` showed zero bytes changed on the DB across the `mutate/restore` cycle).

Corpus-scope floor — added 2026-08-26

The defect

Step 3 passes `--issues` and `--fixed` to `workable-items diff`, and those are the **only two** markdown renderings it inspects. The tracker artifact is **four** files, not two: the binary's `export` subcommand also writes

Issues_Summary.md and Fixed_Summary.md beside them. Those two were outside the gate's corpus entirely, so they could carry arbitrary text while the gate still reported "DB ↔ Markdown in sync".

Measured, with a decisive control:

Tree	Gate output	Exit
Real DB, real Issues.md/Fixed.md, both _Summary files replaced with "TOTALS ARE FABRICATED: 9999 open items, none of which exist."	diff: DB and Markdown are in sync	0
Same tree, one byte appended to Issues.md instead	~ LVA-152 body differs ... diff: 1 difference(s)	1

Fabricating half the artifact passed; touching the other half failed. A gate that certifies "in sync" having read half the artifact asserts nothing about the other half — and the _Summary files are precisely the ones a reader skims for totals.

Why the check is done here rather than in the binary

The canonical binary's diff subcommand takes no _Summary flags — workable-items diff --help offers only -db / -issues / -fixed. Rather than reimplement the summary format on the Lava side (which would drift from the renderer the moment either changes), the gate **regenerates** the summaries from the same DB with export --no-formats into a temp dir and compares bytes. Same source of truth, same renderer, no second implementation to keep in lockstep.

What the gate now refuses

Three refusals, all **exit 1**:

1. **A _Summary file is absent** — summary tracker(s) ABSENT — the gate would certify a partial artifact. The message states what was examined (2 of the 4 files 'export' produces) and distinguishes the cause: these are *output* files of the same renderer, so an absent one is a regeneration that never ran or was reverted, not an optional artifact. The remedy printed is the exact export --no-formats command.
2. **Regeneration failed** — could not regenerate the summary trackers for comparison. Explicitly labelled a **tooling failure, not**

- a sync verdict:** skipping it would let the summary half go unchecked in silence. The binary's stderr is reproduced, indented.
3. **A _Summary file is stale** — summary tracker(s) are STALE against the DB, with the drifting paths named and the first differing lines of a unified diff shown so the drift is visible without a second command.

Paths and overrides

Issues_Summary.md is resolved beside `--issues`, Fixed_Summary.md beside `--fixed`. Both can be overridden for hermetic testing via `LAVA_WORKABLE_ITEMS_ISSUES_SUMMARY` and `LAVA_WORKABLE_ITEMS_FIXED_SUMMARY`. The temp dir used for regeneration is removed on exit via `trap`; the committed files are only ever **read**, never rewritten by the gate — it reports drift, it does not silently repair it.

§6.J falsifiability rehearsal of the floor itself

1. Overwrite docs/Issues_Summary.md with arbitrary text → run `bash scripts/check-workable-items.sh` → expect **exit 1**, summary tracker(s) are STALE against the DB, docs/Issues_Summary.md named, and the differing lines shown. Before this change: exit 0, DB ↔ Markdown in sync.
2. `rm docs/Fixed_Summary.md` → expect **exit 1**, summary tracker(s) ABSENT.
3. Regenerate with the printed export `--no-formats` command → expect exit 0 and the four-tracker success line.

Usage

```
bash scripts/check-workable-items.sh
```

To regenerate the trackers after editing the DB:

```
constitution/scripts/workable-items/bin/workable-items sync db-to-  
md \  
--db docs/workable_items.db --out-issues docs/Issues.md --out-  
fixed docs/Fixed.md
```

Exit codes

- 0 — DB validated, **all four** rendered trackers (`Issues.md`, `Fixed.md`, `Issues_Summary.md`, `Fixed_Summary.md`) in sync with the DB, DB tracked. The success line names all four.
- 1 — validation failure, divergence in any of the four trackers, a missing `_Summary` file, a failed regeneration, build failure, or DB untracked.

Hermetic test

`tests/check-workable-items/test_workable_items_sync.sh` — positive case (clean tree PASS), negative case (corrupt docs/`Issues.md` → gate FAILS), restore → PASS.

Inheritance

Classification: project-specific (the LVA key + Lava's doc set are Lava-specific; the §11.4.93/95 mandates are universal per HelixConstitution).